



UX Standards and Accessibility

TASCO Insurance × VETC Growth Platform — User Experience

iorta TechNXT

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These standards apply to all three v1 surfaces: the staff console (/), the customer app (/app/) and the certificate check (/verify/<certNo>). They are part of the Definition of Done (see the Delivery Methodology, docs/delivery/delivery-methodology.pdf).

Conformance target: WCAG 2.2 Level AA on all surfaces. This also serves Vietnam's accessibility expectations for public-facing digital services and the VETC app's own webview standards.

1. WCAG 2.2 AA — requirements and how we apply them

1.1 Perceivable

SC	Requirement	How applied
1.1.1 Non-text content	Text alternatives	Logos have alt="TASCO Insurance". Decorative icons have aria-hidden="true". The QR code has alt text containing the certificate number and the verification URL in text below it. Charts have a data-table alternative.
1.3.1 Info and relationships	Semantic structure	Real <table> with <th scope>. Form <label for>. Headings in order (one h1 per page). Lists for navigation.
1.3.2 Meaningful sequence	DOM order = visual order	No CSS reordering of interactive content.
1.3.4 Orientation	Not locked	Customer app works in portrait and landscape.
1.3.5 Identify input purpose	autocomplete	username, current-password, one-time-code (MFA), tel.
1.4.1 Use of colour	Never colour alone	Tier, status and confidence badges always have text. Errors have an icon and text.
1.4.3 Contrast (minimum)	4.5:1 text, 3:1 large	Token pairs verified in light and dark (Design System, section 3.1). Teal #70CAC8 is never used for text on white.
1.4.4 Resize text	200 % without loss	rem-based type. Layouts reflow.
1.4.10 Reflow	320 CSS px without horizontal scroll	Tables collapse to cards. Navigation collapses. The Rules studio JSON editor scrolls inside its own container only.
1.4.11 Non-text contrast	3:1 for UI components	Input borders on focus or hover, the focus ring (--focus #1E6FFF), toggle tracks, chart lines.
1.4.12 Text spacing	Survives user spacing overrides	No fixed-height text containers.
1.4.13 Content on hover or focus	Dismissible, hoverable, persistent	Tooltips (score reasons) open on focus and hover and close with Esc.

1.2 Operable

SC	Requirement	How applied
2.1.1 Keyboard	Everything by keyboard	All actions reachable with Tab/Shift+Tab, Enter/Space. Custom widgets follow ARIA Authoring Practices patterns. Row "Open" links.

SC	Requirement	How applied
2.1.2 No keyboard trap		Drawers and modals trap focus intentionally but close with Esc and return focus.
2.1.4 Character key shortcuts	Can be turned off or need a modifier	Shortcuts (section 1.5) only fire when focus is not in an input, and can be turned off in the user preferences.
2.2.1 Timing adjustable	Session timeout warning	Staff JWT lifetime is 30 min (JWT_TTL_SECONDS). The console warns 2 min before expiry with a "Stay signed in" option, and on expiry shows "Phiên đã hết hạn — vui lòng đăng nhập lại" while keeping unsent form input. The MFA step token lasts 5 min. Quotes are valid for 24 h, and the expiry time is shown on the quote.
2.3.1 Three flashes	None	No flashing content.
2.4.1 Bypass blocks	Skip link	"Bỏ qua đến nội dung chính / Skip to content" is the first focusable element and targets <code><main id="main" tabindex="-1"></code> .
2.4.2 Page titled	<code><title></code> per view	"Hộp việc telesales — TASCO Growth Platform". Updated on route change (hash routes).
2.4.3 Focus order	Logical	On route change, focus moves to the page h1.
2.4.6 Headings and labels	Descriptive	
2.4.7 Focus visible	Always	2 px <code>--focus</code> outline with 2 px offset. Never outline: none without a replacement.
2.4.11 Focus not obscured (min) (new in 2.2)	Sticky header and toasts never hide the focused element	<code>scroll-padding-top</code> equals the header height. Toasts sit in a corner away from the focus path.
2.5.3 Label in name	Visible label is part of the accessible name	
2.5.7 Dragging movements (new in 2.2)	Single-pointer alternative	No drag-only interaction. Assignment uses a select, not drag-and-drop.
2.5.8 Target size (min) (new in 2.2)	$\geq 24 \times 24$ px	Customer app 44 px (<code>--tap</code>). Staff console ≥ 32 px with spacing.

1.3 Understandable

SC	Requirement	How applied
3.1.1 / 3.1.2 Language	<code>lang</code> set	<code><html lang="vi"></code> or "en" follows the language toggle. Mixed-language fragments (English rule IDs) are marked <code>lang="en"</code> .
3.2.1 / 3.2.2 Predictable	No context change on focus or input	Filters apply on "Áp dụng" or after a debounced change with a live-region announcement. Never navigates on select.
3.2.6 Consistent help (new in 2.2)	Help in the same place on every page	"?" in the top bar opens the contextual help drawer on every page. The customer app has "Trợ giúp" in the Tài khoản tab and the hotline in the footer.
3.3.1 / 3.3.3 Error identification and suggestion	Clear errors with a fix	Section 9 patterns. Server validation <code>details</code> mapped to fields.
3.3.2 Labels or instructions	Visible labels and format hints	"Ví dụ: 30A-123.45", "dd/mm/yyyy".

SC	Requirement	How applied
3.3.4 Error prevention (legal, financial)	Review and confirm	Purchase shows a review step (products, premium incl. VAT, period, wallet) before "Thanh toán". DSAR erase and key revocation need confirmation. Rule approval shows the diff and needs a comment.
3.3.7 Redundant entry (new in 2.2)	Don't ask twice	The plate and vehicle are pre-filled from the session. The quote carries into payment. Expiry confirmed once is reused.
3.3.8 Accessible authentication (min) (new in 2.2)	No cognitive test	Password managers and paste allowed. The TOTP field allows paste and <code>autocomplete="one-time-code"</code> . MFA enrolment offers a QR code with a typed-key alternative ("Cannot scan? Enter the key manually"). No CAPTCHA puzzles.

1.4 Robust

SC	Requirement	How applied
4.1.2 Name, role, value	Native elements first. ARIA only where needed.	Toggles are <code><button aria-pressed></code> . Tabs use <code>role="tablist"</code> . The drawer uses <code>role="dialog"</code> .
4.1.3 Status messages	Live regions	See section 1.6.

1.5 Keyboard map (staff console)

Key	Action
Tab / Shift+Tab	Move focus
Enter / Space	Activate
Esc	Close drawer, modal or menu
? (Shift + /)	Open contextual help (when not typing)
g then i / l / r	Go to Telesales inbox / Leads / Rules studio (optional, can be turned off)
/	Focus the page search or filter
Arrow keys	Move within tabs, menus and radio groups

1.6 ARIA landmarks and live regions

Landmark	Element
Banner	<code><header></code> (top bar)
Navigation	<code><nav aria-label="Main"></code> (left navigation), <code><nav aria-label="Breadcrumb"></code> where used
Main	<code><main id="main"></code>
Complementary	Help drawer <code><aside></code> (dialog when open)
Contentinfo	<code><footer></code> (iorta TechNXT credit, version from <code>/api/meta</code>)

Live region	Politeness	Announces
Toast region	polite (assertive for errors)	Saved, claimed, approved, copy-guard rejection
List status	polite	"Hiện thị 50 / 2.431 khách hàng tiềm năng" after filters
Voice console transcript	polite	New bot line (<code>role="log"</code>)

Live region	Politeness	Announces
Purchase progress	polite	"Đang thanh toán...", "Đã cấp giấy chứng nhận"
Session warning	assertive	Expiry in 2 minutes

1.7 Reduced motion and other preferences

- `prefers-reduced-motion: reduce` sets `--motion` to `0ms`, disables shimmer and replaces the success animation with a static check.
- `prefers-color-scheme` drives the default theme. The user's choice (`data-theme`) overrides it.
- `forced-colors: active` (Windows High Contrast): borders and focus use system colours. Badges keep their text.

1.8 Testing protocol (per story touching UI)

1. Automated: axe-core with zero serious or critical issues, run in CI on core routes.
2. Keyboard-only walkthrough of the changed flow.
3. Screen readers: NVDA + Firefox (Windows, staff) and VoiceOver + Safari iOS / TalkBack + Chrome Android (customer), in Vietnamese voice.
4. Zoom 200 % and 320 px reflow.
5. Contrast check of any new colour pairing in both themes.
6. Release accessibility audit by an external specialist before G3, then annually.

2. Responsive breakpoints

Name	Min width	Staff console	Customer app
xs	0	Not supported for heavy work (read-only usable)	Design target (360 px). Single column, bottom tab bar.
sm	480 px	Stacked cards	Single column, wider cards
md	720 px	Navigation behind the ☰ menu button. User name, roles and region shown in the top bar. Tables become cards.	Centred column, max-width 560 px
lg	1024 px	Left navigation visible (240 px), 1–2 column content. The sign-in screen is split (brand panel + form) from 900 px.	Same
xl	1280 px	Design target. Content max-width 1440 px. Customer 360 in 3 columns.	Same

Page gutters are 16 px on phones and 24–32 px on desktop.

3. Performance budgets

Metric	Customer app (mid-range Android, 4G)	Staff console (office laptop)	Certificate check
Largest Contentful Paint (p75)	≤ 2.0 s	≤ 2.5 s	≤ 1.5 s

Metric	Customer app (mid-range Android, 4G)	Staff console (office laptop)	Certificate check
Interaction to Next Paint (p75)	≤ 200 ms	≤ 200 ms	≤ 200 ms
Cumulative Layout Shift	≤ 0.05	≤ 0.1	≤ 0.05
JS shipped (gzip)	≤ 70 KB	≤ 150 KB	≤ 20 KB
CSS (gzip)	≤ 20 KB	≤ 30 KB	≤ 10 KB
Fonts	≤ 2 files, font-display: swap, Vietnamese subset	same	System fonts only
API p95 (server)	Quote ≤ 300 ms, order ≤ 1.5 s (wallet + issuance)	Lists ≤ 400 ms, Customer 360 ≤ 600 ms	≤ 150 ms
Taps to renew	≤ 3	—	—

The v1 front end has no build step and uses ES modules, so the budgets are achievable without a framework. Static assets under `/assets/` are cached for 24 h and HTML is `no-cache` (`src/adapters/http/app.js`). Performance is checked with Lighthouse CI (front end) and `npm run test:perf` (API).

4. Internationalisation and localisation

Aspect	Standard
Languages	<code>vi</code> (default) and <code>en</code> . The staff console toggle (VI/EN) is stored per browser. The customer app is Vietnamese, with English only where VETC enables it.
String management	All UI strings in the catalogue (<code>public/js/shared/i18n.js</code>), keyed, with no concatenated sentences. Customer message templates and voice lines are rule content (<code>content.messages</code> , <code>content.voicebot</code>) with <code>vi</code> and <code>en</code> variants, governed by maker-checker.
Data values	Not translated (plates, IDs, rule IDs). Product names use <code>name/nameVi</code> . Tariff categories use <code>label/labelVi</code> . Benefits use <code>title/titleVi</code> .
Dates	Display <code>dd/mm/yyyy</code> . Store and transmit ISO <code>yyyy-mm-dd</code> . The time zone is ICT (UTC+7, as <code>timezoneOffsetHours: 7</code> in <code>contact_policy</code>).
Numbers and currency	<code>Intl.NumberFormat('vi-VN')</code> gives 480.700 đ. Voice uses "đồng". English view: 480,700 VND.
Names	Vietnamese full names in one field (family name first). Masking keeps the given name last ("N. V. An").
Plates and phones	Normalised server-side (<code>src/domain/identity.js</code>). Display formats are 30A-123.45 and 0912***678.
Diacritics	Search and voice intent matching ignore diacritics (<code>stripDiacritics</code>), so "gia han" matches "gia hạn".
Text expansion	Allow 30 % growth for English. No truncation of labels in buttons or navigation.
Sorting	Vietnamese collation (<code>Intl.Collator('vi')</code>) for name lists.

5. Personalisation

Mechanism	Behaviour	Source
Role-based navigation	Only pages the user's permissions allow are shown. The server enforces permissions on every API call regardless of what the UI shows.	GET /api/auth/me → permissions, config/security/rbac.json
Role-based home	Users with <code>dashboard:read</code> land on Home (role-relevant KPIs). Others land on their first permitted page: telesales agents → Telesales inbox, claims handlers → Claims, support engineers → Operations.	Console routing (firstAllowedRoute)
Regional scope	Agents see leads, handoffs and customers of their region (ABAC <code>regional_data</code> , <code>agent_own_handoffs</code>). The region shows in the top bar.	config/rules/abac.json
PII masking by purpose	Name and phone masked without <code>profile:read_pii</code> , with a "Đã che" badge. Every Customer 360 view is audited (<code>profile.viewed</code>).	accessPolicy.maskProfile
Next best action (NBA)	One guided button per lead or customer with the reason ("high intent and call consent on file"). It respects DNC, company-owned routing and low data confidence.	config/rules/nba.json
Benefit relevance	Customers see up to 3 benefits (<code>maxShown</code>) ranked by relevance to them (for example, roadside assistance for long-distance drivers, accident cover for 7+ seat vehicles), with a "why" line. Only approved items appear.	config/rules/benefits.json
Saved preferences	Language, theme, table density, dismissed tips, stored per browser (localStorage, wrapped in try/catch).	Front end

6. Contextual help and guided actions

- Help drawer ("?)** on every console page. It holds a page-specific title and 2–4 short paragraphs (`public/js/console/help.js`), plus links to the relevant persona manual section in `docs/manuals/`. It opens with the "?" key or button and closes with Esc.
- Inline "why" affordances:** score reasons, NBA reasons, blocked-send reasons, copy-guard violations, rule validation errors with JSON path (`$.factors[2].value`).
- Guided action buttons:** the NBA button on Leads and Customer 360 (for example "AI voice bot call → telesales handoff", "Ask customer to confirm expiry in-app", "Route to fleet / B2B team"). Each opens the corresponding pre-filled action (start voice session, send link, quote).
- First-run tips:** a dismissible coach mark on the first visit to the Telesales inbox, Rules studio and Customer 360 (max 3 steps, keyboard accessible, never blocking).
- Empty states that teach:** see the Design System, section 4.11.
- Demo mode helpers (demo environments only):** the sign-in page has **no** role picker or shortcut buttons; everyone signs in with their own credentials. In UAT, the MFA step offers **Đăng nhập bằng mã demo (UAT) / Sign in with demo code (UAT)**. The customer app has a demo vehicle picker. These helpers are hidden when `/api/meta` reports `demoMode: false`.

7. Forms and data entry standards

- Single-column forms. Group related fields with `<fieldset>` and `<legend>`.

- Validate on blur and on submit. Never block typing.
- Preserve input on error and on session expiry.
- Destructive or financial actions need a confirmation step (WCAG 3.3.4).
- Show the server's rule message verbatim when it is user-meaningful (for example "Incident date is outside the policy period").

8. Data display standards

- Show **confidence** next to inferred data (expiry, vehicle category) as a badge with percentage.
- Show **provenance** (lineage chip) for golden-record fields.
- Show **as-of** time on dashboards (`asOf` from `/api/dashboard/overview`).
- Long IDs use mono with a copy button. UUID-like IDs are truncated in the middle (`Q-3f2a...9c1d`) with the full value in the accessible name.

9. Error message patterns

The API returns `{ "error": { "code", "message", "details", "requestId" } }`. The UI maps codes to the patterns below. User-facing text is Vietnamese-first, with the request ID shown in small text for support ("Mã hỗ trợ: 7f3c...").

Code (HTTP)	Situation	Staff console message pattern	Customer app message pattern
VALIDATION_FAILED (400)	Bad input	Inline field errors from <code>details</code> , plus a summary "Vui lòng kiểm tra lại các trường được đánh dấu."	"Thông tin chưa hợp lệ: {field}. {hint}"
UNAUTHENTICATED (401)	Bad credentials or expired session	Sign-in: "Tên đăng nhập hoặc mật khẩu không đúng." Session: "Phiên đã hết hạn — vui lòng đăng nhập lại."	"Đường dẫn đã hết hạn hoặc không hợp lệ. Vui lòng mở lại từ ứng dụng VETC."
FORBIDDEN (403)	Missing permission or ABAC denial	"Bạn không có quyền thực hiện thao tác này ({permission}). Liên hệ quản trị viên nếu cần."	"Không thể thực hiện thao tác này."
NOT_FOUND (404)	Missing entity	"Không tìm thấy {đối tượng}. Có thể đã bị xóa hoặc bạn không có quyền xem."	"Không tìm thấy thông tin."
CONFLICT (409)	Concurrent edit (for example, the handoff was updated by someone else)	"Dữ liệu vừa được người khác cập nhật. Tải lại để xem thay đổi mới nhất." with a Reload button	—
BUSINESS_RULE_VIOLATION (422)	Valid input that breaks a rule (quote expired, DNC, maker-checker, invalid transition, active policy blocks erasure)	Show the server message (it is written for users), for example "Maker-checker: you cannot approve your own change" → "Bạn không thể tự duyệt thay đổi của chính mình."	"Báo giá đã hết hạn — vui lòng lấy báo giá mới." with a primary "Lấy báo giá mới"
ACCOUNT_LOCKED (423)	5 failed attempts	"Tài khoản tạm khóa 15 phút do nhập sai nhiều lần."	—

Code (HTTP)	Situation	Staff console message pattern	Customer app message pattern
RATE_LIMITED (429)	Too many requests or sign-ins	"Quá nhiều yêu cầu — vui lòng thử lại sau {Retry-After} giây."	Same, simplified
UPSTREAM_UNAVAILABLE (503)	Wallet, core or messaging down (circuit open)	"Hệ thống đối tác tạm thời gián đoạn ({integration}). Thao tác chưa được thực hiện; vui lòng thử lại."	"Thanh toán tạm thời gián đoạn. Bạn chưa bị trừ tiền . Vui lòng thử lại sau ít phút." Only claim "not charged" when the order status confirms it.
INTERNAL_ERROR (500)	Unexpected	"Đã xảy ra lỗi không mong muốn. Vui lòng thử lại; nếu vẫn lỗi, gửi mã hỗ trợ {requestId} cho bộ phận hỗ trợ."	"Đã có lỗi. Vui lòng thử lại. Mã hỗ trợ: {requestId}"

Principles: say what happened, say what to do next, never blame the user, never show stack traces, always keep the request ID available, and never lose the user's input.